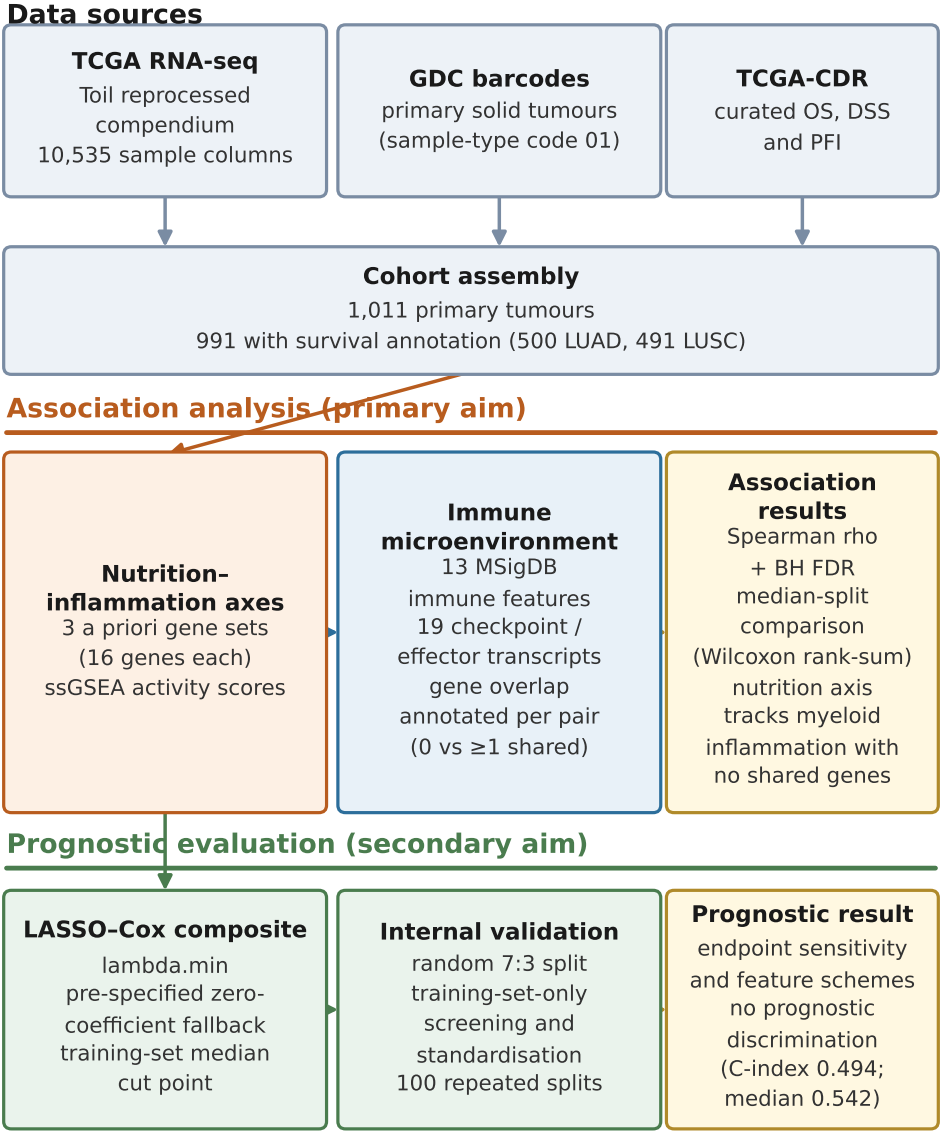
Data sources
TCGA RNA-seq
Toil reprocessed compendium
10,535 sample columns
GDC barcodes
primary solid tumours
(sample-type code 01)
TCGA-CDR
curated OS, DSS and PFI
Cohort assembly
1,011 primary tumours
991 with survival annotation (500 LUAD, 491 LUSC)
Association analysis (primary aim)
Nutrition–inflammation axes
3 a priori gene sets
(16 genes each)
ssGSEA activity scores
Immune microenvironment
13 MSigDB
immune features
19 checkpoint / effector transcripts
gene overlap
annotated per pair
(0 vs ≥1 shared)
Association results
Spearman rho
+ BH FDR
median-split
comparison
(Wilcoxon rank-sum)
nutrition axis
tracks myeloid
inflammation with
no shared genes
Prognostic evaluation (secondary aim)
LASSO–Cox composite
lambda.min
pre-specified zero-
coefficient fallback
training-set median
cut point
Internal validation
random 7:3 split
training-set-only
screening and
standardisation
100 repeated splits
Prognostic result
endpoint sensitivity
and feature schemes
no prognostic
discrimination
(C-index 0.494;
median 0.542)